# BROAD EXAMINATIONS®

# P.7 MATHEMATICS EXAMINATION
# PRE – PLE TRIAL SET I 2023

**Time allowed: 2 hours 15 minutes.**

| Random No. | | | | | | Personal No. | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |

Candidate's Name:.....

Candidate's Signature.....

District Name: .....

**Read the following instructions carefully:**

1. This paper is made up of two sections: **A** and **B**.
2. Section **A** has **20** questions (**40 Marks**)
3. Section **B** has **12** questions (**60 Marks**)
4. Answer **ALL** questions in both sections **A** and **B**.
5. All answers must be written in the space provided in blue or black ball point pens and ink. **Only diagrams should be done in pencil.**
6. Unnecessary crossing of answers will lead to loss of marks.
7. Any handwriting, which cannot be easily read, may lead to loss of marks.
8. Do **not** fill anything in the boxes indicated for Examiners' use only.

| FOR EXAMINERS' USE ONLY | | |
|---|---|---|
| **PAGES** | **MARKS** | **SIGN** |
| Page 2 | | |
| Page 3 | | |
| Page 4 | | |
| Page 5 | | |
| Page 6 | | |
| Page 7 | | |
| Page 8 | | |
| **TOTAL** | | |

**Teacher's comment to the learner**

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*Approved by:*

*Team Head Mathematics Department*

**PLE REVISION TIPS**

**In groups of fives, revise with your friends simple work from P.4 class.**

**Read about:**

- 3- Dimensional figures and their properties
- Tally marks under data handling
- Addition and subtraction of time
- 4 by 4 magic squares and magic triangles

## SECTION .A. (40 Marks)

1. Add: 43 + 16

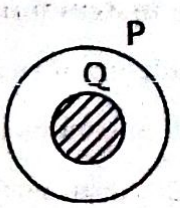
2. Change $13_{five}$ to base ten.

3. Work out the LCM of 12 and 15

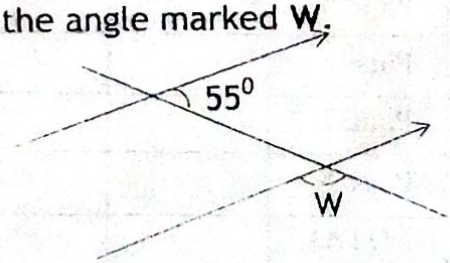
4. Find the range of 3, 7, 8, 5 and 6.

5. A trader sold a dozen picfare books at sh.1200 each book. How much money did the trader get after selling the books?

6. Describe the shaded region in the figure below.




7. In the diagram below, find the size of the angle marked W.




8. What number is increased by 10% to become 22?

9. A driver drove his car at a speed of 20 metres every second. What was his speed in km/h?

10. **Solve:** m + 3 = 1 (mod.7)

13. Write the time 3·50pm in 24 hour clock system.

14. Round off 18.87 to the nearest whole number.

15. If $y = x - 3$, complete the table below.

| | | | |
|---|---|---|---|
| x | 3 | 0 | ___ |
| y | 6 | ___ | 4 |

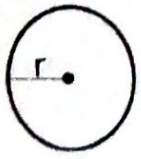
16. Using a ruler, a pencil and a pair of compasses only, construct an angle of $15^0$

17. Express 14 as a product of its prime factors.

18. Set G = {2, 0, 5, 9, 7}. Find the number of proper subsets in set G.

19. Given that $\frac{2}{3}$ of a kilo of maize flour costs sh.1600, find the cost of 4kg of maize flour at the same rate.

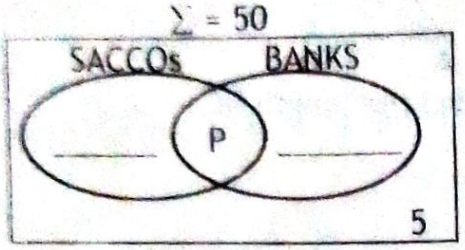
20. The diagram below is a circular end of a cylindrical tin whose circumference is 88 metres. Find its radius.



## SECTION .B. (60 Marks)

21. In a trading centre of 50 traders, 32 have accounts in SACCOs (S), 34 have them in Banks (B), some (P) have them in both institutions while 5 do not have accounts at all.

(a) Represent the above information in the venn diagram below.




(b) How many traders have accounts in both institutions?

(c) If a trader is selected at random to be a business representative, find the probability of choosing one who has an account in SACCOs only.

(05 Marks)

22. (a) **Work out:** $1\ 1\ 0_{two} - 1\ 1_{two}$

(b) If $102p = 201_{five}$, find the unknown base p.

(05 Marks)

23. Study the exchange rates below and answer the questions that follow;

1 United States dollar (US$1) = Ugsh.3500
1 Kenya shilling (Ksh.1) = Ugsh.30
1 British pound Sterling (£.1) = Ugsh.4000

(a) A man has 330 US dollars and Ksh.5500. How much money in Uganda shillings does he have altogether?

(b) A phone costs Ugsh.640,000. Find its cost in pounds.

(05 Marks)


FOR ACADEMIC EXCELLENCE, AVOID MISSING BROAD HOLIDAY PACKAGES, WORK BOOKS & PAST PAPER BOOKLETS

24 The number of pupils in P.1, P.2, P.3 and P.4 at Excel Junior School are 66, 54, 72 and 48 respectively.

(a) Which class has the least number of pupils?

(b) How many more pupils are there in P.3 than in P.1?

(c) Work out the average number of pupils in the four classes.

**(05 Marks)**

25. (a) Using a ruler, a pencil and a pair of compasses only, construct a triangle PLE where line PL = 7cm, angle PLE = $75^0$ and angle EPL = $45^0$

(b) Measure line LE.

**(05 Marks)**

(a) Work out. 1.36 × 2.9

(b) Simplify. 7 × 12.12 ÷ 6.1

**(05 Marks)**

27 Today, Monday 14th September, teachers at Pole pole Primary School have received their salaries. They will receive their next salary after 30 days. On which day and date will they receive their next salary?

**(04 Marks)**

28 **Solve;**

(a) $3a - 5 = 19$

(b) $5k - 3 = k + 5$

**(05 Marks)**

29 A water tank is 80cm long, 60cm wide and 90cm high.

(a) Find its volume.

(b) How much water does it hold when it is $\frac{5}{9}$ full?

**(05 Marks)**

Sarah, Susan, Barbra and Monius have an average age of 22 years. Sarah is n years old, Susan is 5 years older than Sarah, Barbra is 19 years old and Monius is n+4 years old.

Find the value of n.

(b) Find Susan's age.

(05 Marks)

A motorist left town K at 8:30am at a speed of 35km/h. After a 2 hour drive, the car go a problem which took him 30 minutes to solve. He then continued to town L at a speed of 40km/h reaching town L at 12:15pm. He immediately returned to town K at a constar speed of 96km/h.

How far is town L from town K?

(b) Calculate the average speed of the motorist for the whole journey.

(05 Marks)

The pictograph below shows the number of boxes of books received by UPE Schools in a Sub county.

| SCHOOL | NO. OF BOXES. |
|---|---|
| Ntusi | ☒ ☒ ☒ |
| Nzozi | ☒ ☒ ▽ |
| Busega | ☒ ☒ ☒ ☒ |
| Bitaabo | ☒ ☒ ▽ |

Given that ☒ represents 24 books and ▽ represents12 books,

Find the number of books Busega Primary School received.

(b) How many more books did Ntusi get than Bitaabo Primary School?

Work out the total number of books the four schools received.

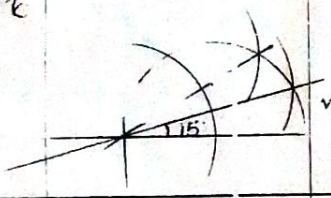
END

(06 Marks)

# BROAD EXAMINATIONS GUIDES

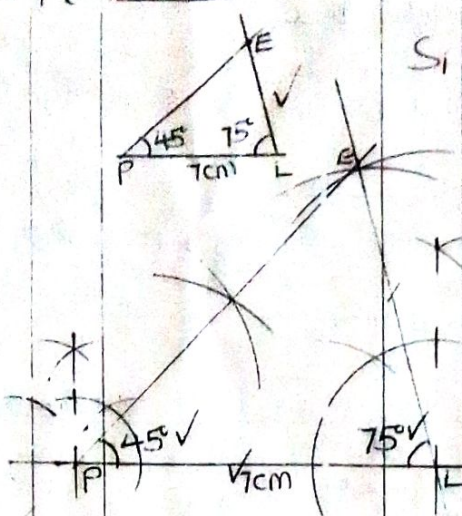
## P.7 MATHEMATICS PRE-PLE TRIAL SET I TERM III 2023

| Qn. | Class | Solution | Marks | Comments |
|---|---|---|---|---|
| 1 | P1 | 43<br>+16<br>59 ✓ | B2 | Cn sight |
| 2. | P5 | (1×5) + (3×5) ✓<br>1×5 + 3×1<br>5 + 3<br>8ten ✓ | M1<br>A1 | for expanding<br>for 8ten |
| 3 | P4 | 2 \| 12 \| 15<br>2 \| 6 \| 15<br>3 \| 3 \| 15<br>5 \| 1 \| 5<br>1 \| 1<br>2×2×3×5<br>4×15 ✓<br>CC ✓ | M1<br>A1 | Accept use of multiples<br>M1 for correct method<br>A1 for CC |
| 4 | P5 | Range = H-L<br>= 8-3 ✓<br>= 5 ✓ | M1<br>A1 | for subtracting<br>for 5 |
| 5 | P4 | Sh.12cc × 12 ✓<br>Sh.144cc ✓ | M1<br>A1 | for multiplying<br>for Sh.144cc |
| 6. | P5 | Cm P or set Sq | B2 | Cn sight |
| 7 | P.6 |  <br>W+55° = 180° (co-int angles) M1<br>W+55°-55° = 180°-55°<br>W = 125° A1 | | for cyl<br>for 125° |
| 8 | P6 | let the no be n<br>11%n + 11% = 11%<br>11/100 × n = 22 M1<br>11n/11 = 22×100/11<br>n = 200 ✓ | M1<br>A1 | for correct method<br>for 200 |
| 9. | P6 | 72km/hr = 72000m/3600sec<br>= 72km/1hr - 1hr/3600sec ✓<br>= 72km × 1hr/3600 ✓<br>= 72km/hr ✓ | M1<br>A1 | for dividing<br>for 72km/h |
| 10 | P7 | n+3 = 1 (mod 7)<br>n+3-3 = 1-3 (mod 7) ✓<br>n = 1-3 (mod 7)<br>n = 4-3 (mod 7)<br>n = 5 (mod 7) ✓ | M1<br>A1 | for subtracting 3<br>for n=5 |
| 11 | P6 | 3x-(x+4)<br>3x-x-4 ✓<br>2x-4 ✓ | M1<br>A1 | for removing brackets |
| 13. | P6 | 3 5L<br>+12 L ✓<br>15 5L hrs | M1 | for adding |

| Qn | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 13 | PE | $\frac{1}{2}sh = \frac{1}{2}bh$<br>$\frac{1}{20} \times \frac{2}{3} \times h = \frac{1}{24} \times 12 \times 24$ ✓<br>$\frac{2h}{2} = \frac{24}{2}$<br>$h = 12cm$ ✓ | M1<br>A1 | for correct method<br>for 12cm |
| 14 | PE | T O th<br>19 ~~12~~<br>+ 1 ✓<br>19 ✓ | M1<br>A1 | for adding 1<br>for 19 |
| 15 | P7 | Value of x<br>$y = x - 3$<br>$4 = x - 3$<br>$4 + 3 = x - 3 + 3$<br>$7 = x$ ✓<br>Value of y<br>$y = x - 3$<br>$y = 0 - 3$<br>$y = -3$ ✓ | B1<br>B1 | for any correct value of y<br>1 and -3 |
| 16 | PE |  <br> ✓ | B1 | for a well constructed angle |
| 17 | PE | 2 \| 14 ✓<br>7 \| 7<br>1<br>$14 = 2 \times 7$ ✓ | M1<br>A1 | for correct method<br>for $2 \times 7$ |
| 18 | PE | $2^5 - 1$<br>$2^5 - 1$ ✓ $M_1$<br>$2 \times 2 \times 2 \times 2 \times 2 - 1$<br>$32 - 1$<br>31 proper subsets $A_1$ | | for $-1$<br>for 31 |
| 19 | PE | $\frac{2}{3}$ kg → sh 160<br>1 kg → sh 160 ÷ 2/3<br>→ sh ~~480~~ 240 × 3/2<br>→ sh 240<br>4 kg → sh 240 ✓ M1<br>× 4<br>sh 960 ✓ A1 | M1<br>A1 | for multiplying<br>for sh 960 |
| 20 | PE | $2\pi r = C$<br>$2 \times \frac{22}{7} r = 88$ ✓ M1<br>$\frac{1}{44} \times \frac{44}{7} r = \frac{2}{88} \times \frac{7}{44}$<br>$r = 14$ metres A1<br>or<br>$\pi d = C$<br>$7 \times \frac{22}{7} d = 88 \times 7$ ✓ M1<br>$\frac{22d}{22} = \frac{88 \times 7}{22}$<br>$d = 28$<br>radius = 14 m ✓ A1 | M1<br>A1 | for correct substitution<br>for correct method<br>for correct |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 21 | P6 (a) | $\xi = 50$<br>SACCOS = 32, BANKS = 34<br>$32 - P$ ✓, P, $34 - P$ ✓<br>5 | B1<br>B1 | for $32 - P$<br>for $34 - P$ |
| | (b) | $(32 + 34 + 5) - 50$ ✓<br>$71 - 50$<br>21 traders ✓ | M1<br>A1 | for a correct method<br>for 11 |
| | (c) | n(s) only<br>$32 - 21$<br>11 ✓<br>Prob = $\frac{11}{50}$ ✓ | B1 | for $\frac{21}{50}$ |
| 22 (a) | P7 | $^{0}\overset{8}{X}\overset{2}{X}\emptyset_{two}$<br>$-\ 11_{two}$<br>$11_{two}$ ✓ B2 | 05 | on sight |
| (b) | | $102_p = 201_{five}$<br>$(1 \times p^2) + (2 \times p^1) = (2 \times 5^2) + (1 \times 5^0)$ ✓ M1 for expanding<br>$p^2 + (2 \times 1) = 2 \times 5 \times 5 + (1 \times 1)$<br>$p^2 + 2 = 50 + 1$<br>$p^2 + 2 - 2 = 51 - 2$ ✓ M1 for subtracting<br>$\sqrt{p^2} = \sqrt{49}$<br>$p = 7$<br>$p =$ seven ✓ A1 for 7 | | |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 23 | P6 | (a) US dollars to Ugsh<br>330 × Ugsh 3500<br>Ugsh 1155000 ✓<br>Ksh to Ugsh<br>5500 × Ugsh 30<br>Ugsh 165000 ✓<br>TOTAL<br>Ugsh 1155000<br>+ Ugsh 165000<br>Ugsh 1320000 ✓<br>(b) Ugsh ~~64,000~~<br>Ugsh ~~4000~~<br>(05) 160 pounds ✓ | B1<br>B1<br>B1<br>M1<br>A1 | for Ugsh 1155000<br>for Ugsh 165000<br>for Ugsh 1320000<br>for dividing<br>for 160 |
| 24 | P5 | (a) P4 ✓<br>(b) ~~72~~ 66<br>$-\ 66$<br>6 more pupils<br>(c) $66 + 54 + 72 + 48$<br>4<br>~~240~~ 240<br>60<br>60 pupils ✓ | B1<br>M1<br>A1<br>M1<br>A1 | for subtracting<br>for 6<br>for adding and dividing<br>for 60 |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 25 | PC | Sketch   | $S_1$ | for a detailed sketch |
| | | $C_1$ for 45° | | |
| | | $C_1$ for 75° | | |
| | | $L_1$ for 7cm | | |
| | | 45°✓ 75°✓ P 7cm L | | |
| | | (b) LE = 5.9 ± 1 cm ✓ | 05 | $B_1$ |
| 26 | P5 | (a) $\overset{3}{\cancel{4}}\overset{18}{\cancel{8}}E$ ✓<br>$-2.9$<br>$1.46$ ✓ | M1<br>A1 | for arranging according to PV<br>for 1.46 |
| | | (b) $(7+6.3)-12.12$<br>$7.0$<br>$+6.3$<br>$13.3$ ✓ | B1 | for 13.3 |
| | | $13.\overset{2}{3}\overset{10}{0}$<br>$-12.12$ ✓<br>$1.18$ ✓ | M1<br>A1 | for subtracting<br>for 1.18 |
| | | | 05 | |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 27 | P7 | $1+30 = __ \pmod 7$ ✓ | M1 | |
| | | $31 = __ \pmod 7$ | | |
| | | $31-7 = 4 \text{ r } 3$ | | |
| | | $31 = 3 \pmod 7$ ✓ | A1 | |
| | | The day will be Wednesday | ✓B1 | for correct day |
| | | Days remaining in Sept.<br>$30-14 = 16$ | | |
| | | Days in Oct.<br>$30-16 = 14$ | | |
| | | The date will be Wednesday, 14th October ✓ | B1 | for correct date |
| | | | 04 | |
| 28 | P6 | (a) $3a-5+5 = 19+5$ ✓ | M1 | for collecting like terms |
| | | $\frac{3a}{3} = \frac{24}{3}$ | | |
| | | $a = 8$ ✓ | A1 | for $a = 8$ |
| | | (b) $5k-3 = k+5$ | | |
| | | $5k-k = 5+3$ ✓ | M1 | for collecting like terms |
| | | $\frac{4k}{4} = \frac{8}{4}$ ✓ | M1 | for dividing by 4 |
| | | $k = 2$ ✓ | A1 | for $k = 2$ |
| | | | 05 | |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 29 | P6 | (a) $80\text{cm} \times 60\text{cm} \times 90\text{cm}$ ✓ | M1 | Emphasize correct units |
| | | $4800\text{cm}^2 \times 76\text{cm}$ | | |
| | | $432000\text{cm}^3$ ✓ | A1 | |
| | | (b) $\frac{5}{9_1} \times \overset{4800}{\cancel{43200}} = \cancel{5}\text{cm}^3$ | | |
| | | $5 \times 4800\text{cm}^3$ | | |
| | | $24000\text{cm}^3$ ✓ | B1 | |
| | | Water in litres<br>$\frac{24000\text{cm}^3}{1000\text{cm}^3}$ ✓ | M1 | |
| | | 24 litres ✓ | A1 | |
| 30 | P6 | (a) Total age = $22 \times 4$ | | |
| | | $= 88$ yrs ✓ | $P_1$ | for sum |
| | | Sarah — n<br>Susan — n+5<br>Nicnius — n+4<br>Barbra — n+4+19<br>— n+23 | | |
| | | $n+n+5+n+4+n+23 = 88$ ✓ | M1 | for eqn |
| | | $4n+32-32 = 88-32$ ✓ | M1 | |
| | | $\frac{4n}{4} = \frac{56}{4}$ | | |
| | | $n = 14$ ✓ | A1 | for 14 |
| | | (b) Susan → $14+5 = 14+5$ | | |
| | | $= 19$ yrs ✓ | $P_1$ | |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 31 | P7 | (a) Dist covered before problem<br>$\frac{35\text{km}}{1\text{hr}} \times 2\text{hr} = 70\text{km}$ ✓ | P1 | for 70km |
| | | Total time from K to L<br>$\overset{11}{\cancel{12}}\overset{75}{\cancel{15}}$<br>$-8.31$<br>$3.45 = 3\frac{\overset{3}{\cancel{15}}}{\cancel{60}_4}$ | | for correct time |
| | | $= 3\frac{3}{4}\text{hrs}$ | | |
| | | Time taken after the problem<br>$3\frac{3}{4} - (2\text{hrs} + \frac{1}{2}\text{hr})$ | | |
| | | $3\frac{3}{4} - 2\frac{1}{2}$ | | |
| | | $3.45$<br>$-2.30$<br>$1.15 = 1\frac{1}{4}\text{hrs}$ ✓ | | |
| | | Distance covered after rest<br>$40 \times 1\frac{1}{4}$ | | |
| | | $\overset{10}{\cancel{40}} \times \frac{5}{\cancel{4}_1} = 50\text{km}$ ✓ | $P_1$ | for correct |
| | | Total distance<br>$70\text{km} + 50\text{km}$<br>$120\text{km}$ ✓ | $B_1$ | for correct |

| Qn. | Class | Solution | Mrks | Comments |
|---|---|---|---|---|
| 31 | P7 | (b) Total distance covered<br>120 km + 120 km<br>= 240 km ✓<br>Time taken returning<br>$\frac{\cancel{120}^{10}}{\cancel{96}_{8}}$ hrs<br>$\frac{5}{8_{4}}$ hrs<br>$1\frac{1}{4}$ hrs ✓ | B1 | for $1\frac{1}{4}$ hrs |
| | | Total time taken<br>$3\frac{3}{4} + 1\frac{1}{4} = 3 + 1 + \frac{3}{4} + \frac{1}{4}$<br>$= 4 + 1$<br>$= 5$ hrs | | |
| | (05) | Av S = $\frac{\cancel{240}^{48}\text{ km}}{\cancel{5}_{1}\text{ hrs}}$<br>= 48 km/hr ✓ | B1 | for 48 km/hr |
| 32 | P5 | a) $24 \times 4$ ✓ = 96 ✓ books | M1<br>A1 | for multiplying or adding |
| | | b) $3 - 2\frac{1}{2} = \frac{1}{2}$<br>$\frac{1}{\cancel{2}_{1}} \times \cancel{24}^{12}$ ✓ = 12 ✓ more books | M1<br>A1 | for multiplying |
| | | c) 24 × 12 ✓<br>288 books | M1<br>A1 | for multiplying |